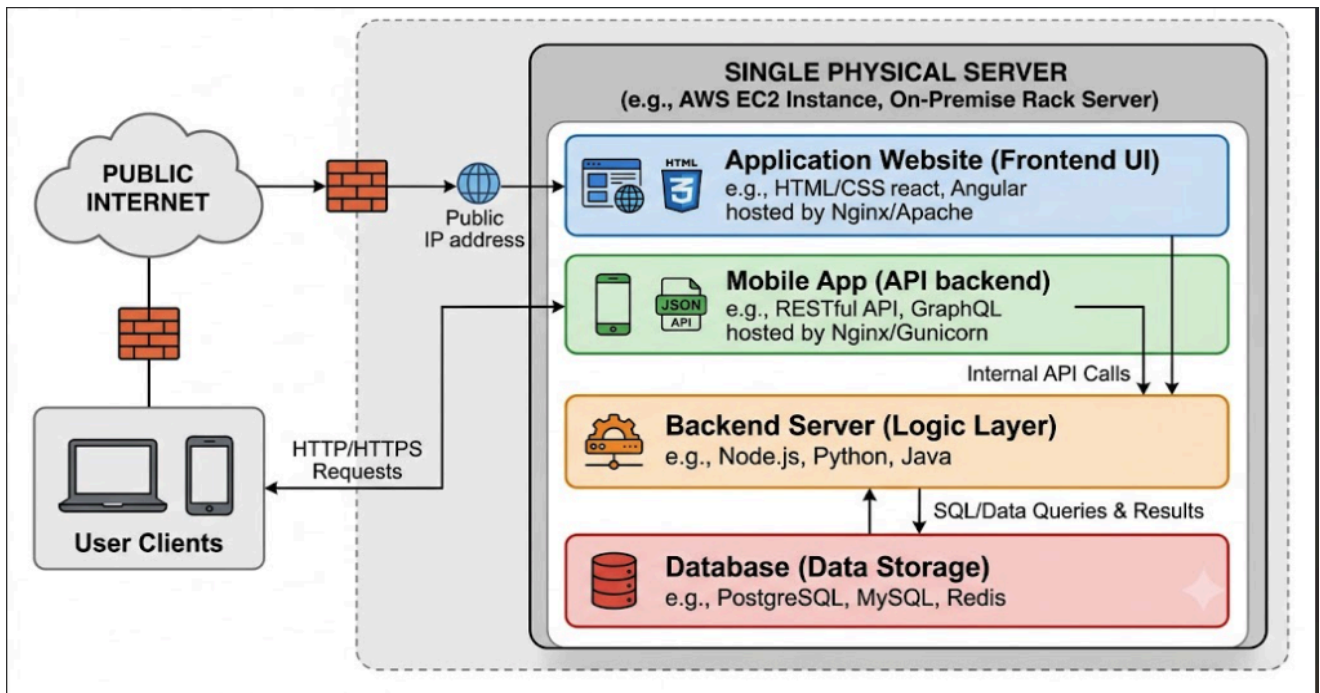


Single Server Setup

- Application website, mobile app, backend server, database all reside in the same server



Advantages

- Cost-Effective & Simple Setup
- Low Latency for Internal Calls
- Easier Maintenance & Debugging

Disadvantages

- Single Point of Failure (SPOF)
- Resource Contention
- Limited Scalability

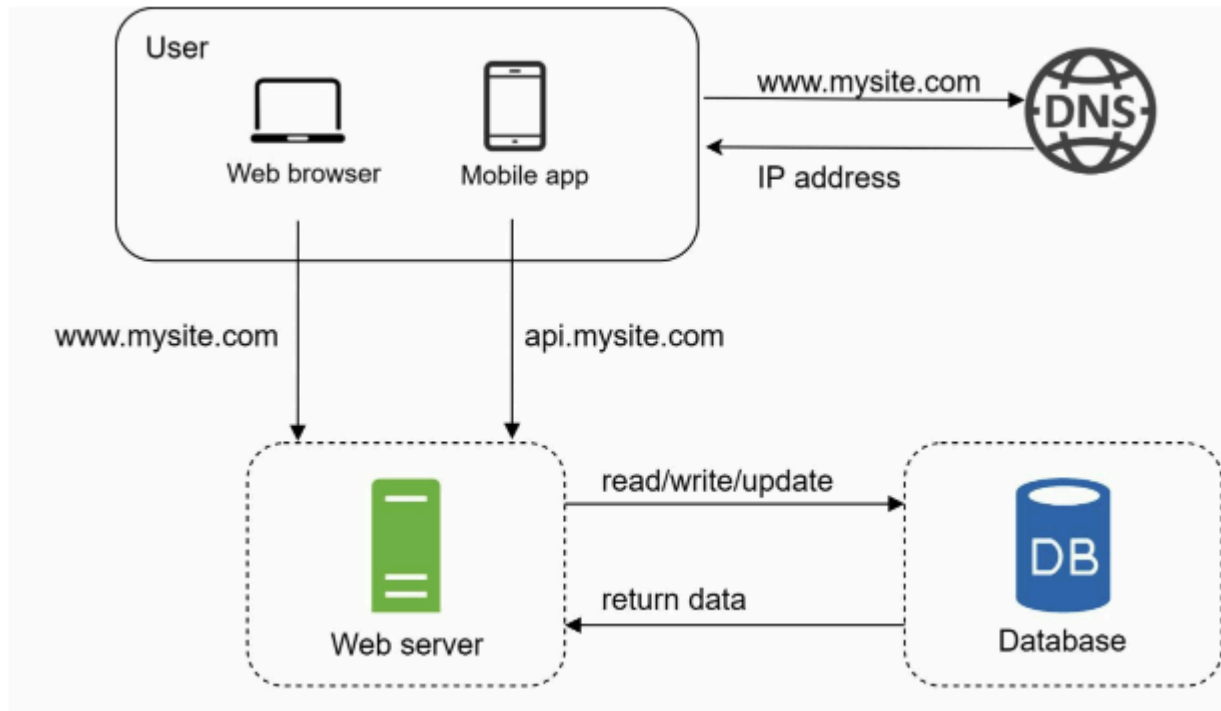
Use Case 1:

The application is write-intensive, processing large multimedia uploads and frequent data modifications. Data retrieval requests are minimal, and the backend server is capable of efficiently handling request processing. The primary workload is driven by high-volume data ingestion and updates.

Any thoughts...? How to solve this?

- VERTICAL SCALING?
- HORIZONTAL SCALING?

Solution: Have a dedicated database server!



Benefits:

- improved Performance
- Better Scalability
- Enhanced Reliability and Maintenance

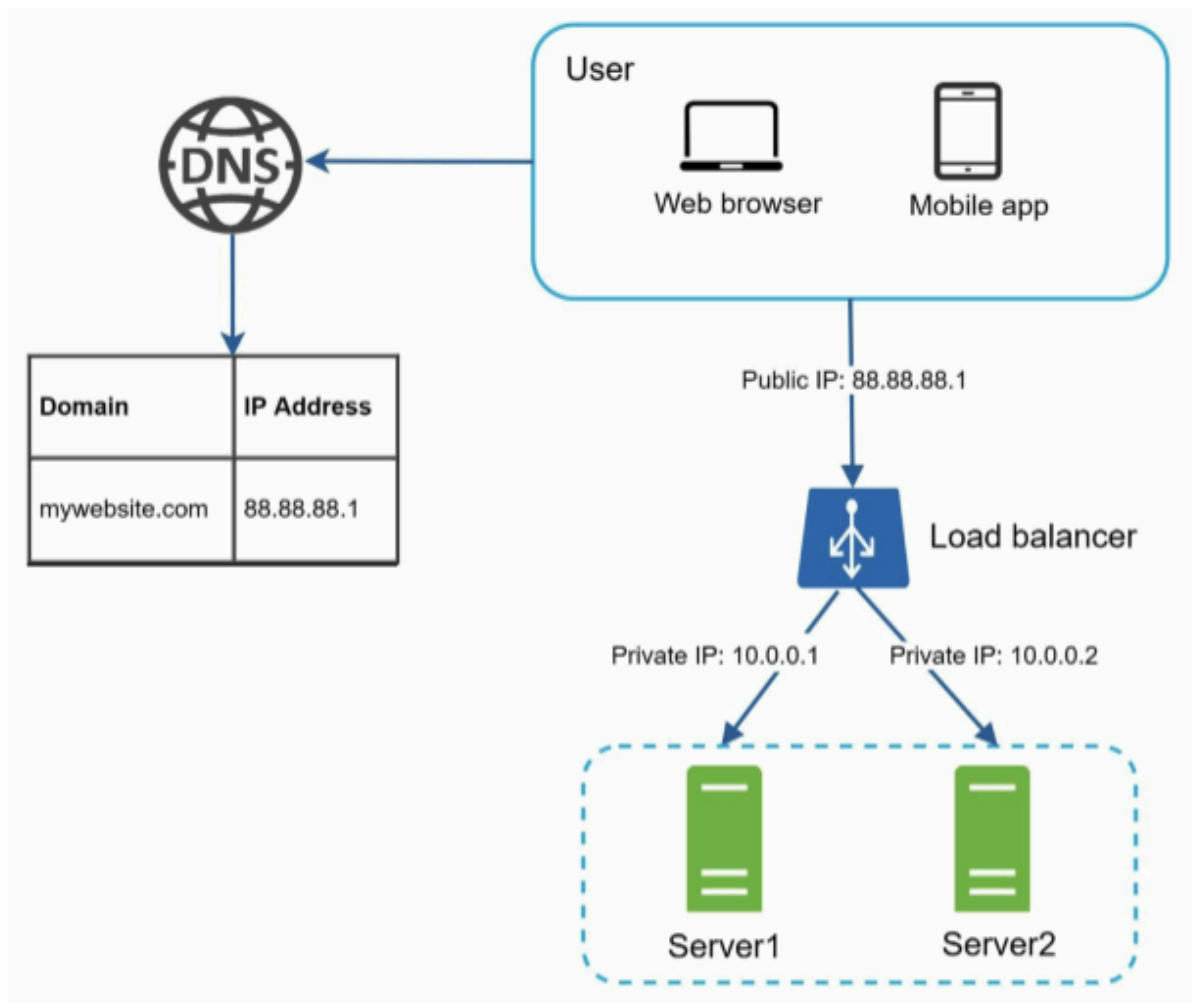
Use Case 2

The application experiences a significant increase in traffic across upload, retrieval, and update operations.

As request volume continues to grow, the backend web server becomes overloaded and is unable to process incoming requests efficiently, resulting in increased response times and reduced application performance.

Any thoughts...? How to solve this?

Solution - Load balancer!

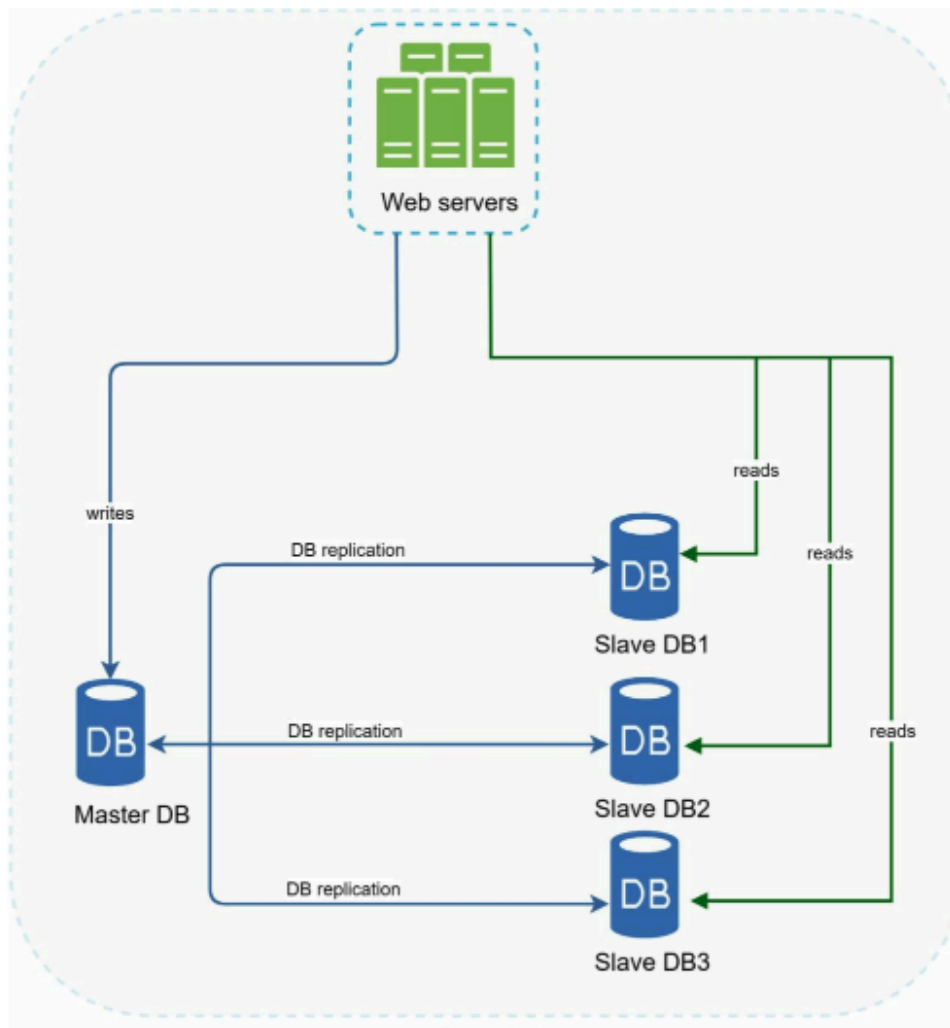


Benefits:

1. Improved Scalability
2. High Availability and Fault Tolerance
3. Better Performance and Resource Utilization

USE CASE 3

- The application is experiencing a very high volume of concurrent upload, retrieval, and update requests. The load balancer and multiple backend servers are able to handle incoming traffic efficiently; **however, the database has become the primary performance bottleneck**. Since it must process all read and write operations, the database experiences high resource utilization, **resulting in slower query execution**, increased response times, and reduced overall application performance.



Benefits

- Improved Database Performance
- Higher Availability for Read Operations
- Serving all concurrent requests

